Year 7 Mathematics
Algebraic Techniques Practice Test 1

Name Array Bustogi

1. a) List the individual terms in the expression 5a + b + 14c . 5a , b , 14c

   b) State the coefficient of each pronumeral in the expression 5a + b + 14c . 5, 1, 14

   c) Give an example of an expression with exactly two terms, one of which is a constant term. $7xy^2+5$

2 Write an expression for each of the following.

   a) 3 more than a $3+a$

   b) 5 less than m $m-5$

   c) the sum of a and b $a+b$

   d) double the value of y $2y$

   e) the product of a and b ab

3. Write an expression for each of the following without using the x or ÷ symbols.

   a) a is halved, then 6 is added $\frac{a}{2}+6$

   b) the sum of a and b is taken and then divided by 3 $\frac{a+b}{3}$

   c) the sum of a and one-third of b $a+\frac{b}{3}$

   d) 7 is subtracted from y and the result is tripled $3(7-y)$

4 Given that a = 5, evaluate:

   a) a + 7 = 12 b) 8a = 40 c) $\frac{10}{a}+4-a = 6-a$

5 Substitute x = 4 and y = 7 to evaluate these expressions.

   a) 5x + y + 8 = 35 ($20+7+8=35$) b) 80 – (2xy + y) = 17 ($80-56+7=17$)

6 If p = 4 and t = 5, find the value of:

   a) $3p^2 = 144$ (3×4^2) b) $t^2 + p^3 = 89$ ($25+64=89$) c) $\sqrt{p^2+3^2} = \sqrt{16+9} = \sqrt{25} = 5$

   $16+9=25$, $25+64=89$

7 Which two of these expressions are equivalent: 3x + 4, 8 – x , 2x + 4 + x ? $3x+4$ & $2x+4+x$

8 Which of the following pairs are like terms? Select all that apply

   a) 3a and 2a b) 3b and 3c c) 2ab and 5ba d) 4k and k e) 2a and 4ab

9 Simplify the following by collecting like terms.

a) $7b + 2 + 3b = 10b + 2$

b) $12d - 4d + d = 8d + d = 9d$

c) $5 + 12a + 4b - 2 - 3a = 5 + 12a + 4b - 2 - 3a = 9a + 4b + 3$

d) $13a + 8b + 2a - 5b - 4a = 13a + 8b + 2a - 5b - 4a = 11a + 3b$

e) $12uv + 7v - 3vu + 3v = 12uv + 7v - 3vu + 3v = 9uv + 10v$

10 Write 4 x a x b x c without multiplication signs. $4abc$

11 Simplify 4a x 2b x 3c, giving your fi nal answer without multiplication signs. $24abc$

12 Simplify 3w x 4w. $12w$

13 Write $(2a + 1) \div 3$ without a division sign. $\frac{2a+1}{3}$

14 Simplify the expression $\frac{8ab}{12b} = \frac{8a}{12} = \frac{4a}{6} = \frac{2a}{3}$

15 Expand the following expressions.

a) $5(x + 3) = 5x + 15$

b) $8(a - 4) = 8a - 32$

c) $3(a + 2b) = 3a + 6b$

d) $5a(3p - 7q) = 15ap - 35aq$

Answers

1. a) 5a, b, 14c .

   b) coefficient of a is 5 coefficient of b is 1 coefficient of c is 14 .

   c) An example is 2a + 7 you could a different example – check

2. a) a + 3

   b) m – 5

   c) a + b

   d) 2 x y

   e) a x b

3. a) $\frac{a}{2}+6$

   b) $\frac{a+b}{3}$

   c) $a+\frac{b}{3}$

   d) 3(y – 7)

4 a) 12 b) 40 c) 1

5 a) 35 b) 17

6 a) 48 b) 89 c) 5

7 3x + 4 and 2x + 4 + x ?

8 a) yes b) no c) yes d) yes e) no

9 a) 10b + 2

   b) 9d

   c) 5 + 9a + 4b + 3

   d) 11a + 3b

   e) 9uv + 10v

10 4abc

11 24abc

12 $12w^2$.

13 $\frac{2a+1}{3}$

14 Simplify the expression $\frac{2a}{3}$

15 Expand the following expressions.

a) $5x + 15$

b) $8a - 32$

c) $3a + 6b$

d) $15ap - 35aq$